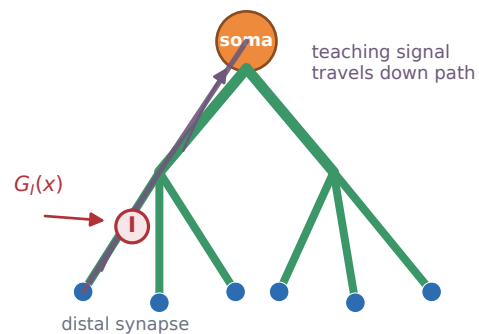


A Path gain is conductance-gated

$G_I(x)$ lowers R_k^{tot} and attenuates all upstream paths

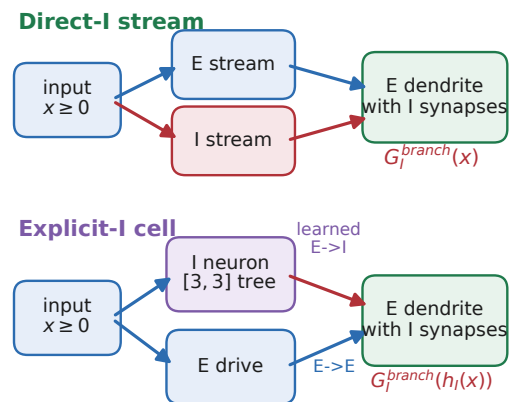


$$\alpha_n(x) = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} R_k^{tot}(x) g_{i \rightarrow k}^{den}$$

$\partial \log \alpha_n / \partial G_k^I = -R_k^{tot}$ on the path

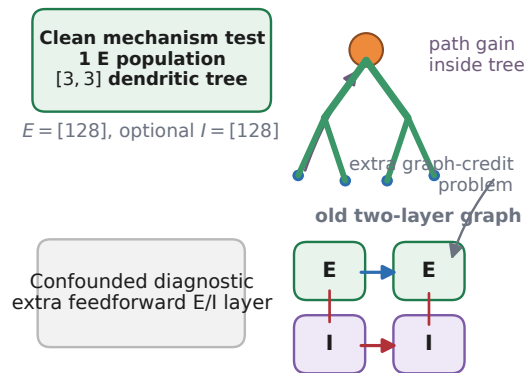
B Two sources, same branch conductance

Both routes end as nonnegative I-to-E conductance on the dendrite



C Clean probe: dendritic depth, not network depth

The path-gain question needs a branched tree, not two feedforward layers



D

One-layer probe result

